Historical and Conditional Forecasts of Sampled Net-Position Peaks

A retrospective benchmark on an FX record and Treasury position aggregates

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Abstract

We benchmark established forecasts of the 0.95 quantile of the largest absolute net position observed at issue and over the next H source records. Historical constructions are evaluated against ridge quantile regression (QR) and local quantiles using completed responses and matched-history controls. The FX evaluation archive had previously been inspected; the Treasury exercise replays a downloaded vintage by economic date rather than historical public release. On 6,625 FX origins, the declared six-record-horizon QR primary lowers mean pinball loss by 12.66% against its original rolling reference and 14.77% against a matched-history reference, with 376 breaches versus 332 and 360. On 277 weekly Treasury-aggregate origins, the separately fitted six-week QR primary raises loss by 50.79% and breaches from 28 to 54. Exploratory AR replay improves on matched increment translation in five of eight cells while breaches increase in all eight. Reweighting eight archived .95 reports distinguishes the FX short-horizon reductions in both mean error components from the small 24-record gain, which reverses above an underprediction/overprediction weight ratio of 20.63, near the stipulated 19. These retrospective comparisons establish source-, horizon-, history- and family-dependent rankings, without coverage guarantees or observed trading and funding benefits.

1. Introduction

A terminal position forecast can miss a large intermediate magnitude that subsequently recedes. We study the largest absolute source-reported net position at issue and over the next H recorded observations. Its upper-quantile report provides a level against which subsequent recorded magnitudes can be monitored. Sampling and netting matter: the target does not reconstruct unobserved inter-record excursions, a complete hedged dealer book or a cash-funding requirement.

The empirical question is whether simple historical peak constructions remain competitive with established regression and local quantile methods fitted directly to this response. We compare historical headrooms (F), increment paths reanchored to the current signed position (S), and native historical peaks subject to the current magnitude floor (K). These historical rules already use the current magnitude or signed anchor. We use direct conditional methods as a shorthand for ridge quantile regression (QR), multivariate nearest-neighbor quantiles (KNN) and a scale-adjusted local variant. Earlier one-state retrieval remains a reference. The methods are evaluated on common issue origins, with controls matching completed-response histories and refit schedules.

The two declared primaries have opposite directions. At the six-record FX horizon, QR with 2,880 completed episodes lowers mean pinball loss by 12.66% against the original rolling 720-episode F rule and by 14.77% against matched-history F. Breaches increase to 376 from 332 and 360, respectively, while mean underprediction amount falls. The separately fitted six-week Treasury QR primary raises loss by 50.79% and increases breaches from 28 to 54 over 277 origins. All four Treasury QR cells lose to their own F controls. The weekly Treasury aggregate measures a different financial object from the individual-market-maker FX record; the two source primaries do not estimate a common replicated effect.

The contribution is a reproducible comparison on a common sampled-peak target. Direct regression and local quantiles can change the ranking of historical constructions even when completed-label histories and update schedules are matched. The favorable short-horizon FX comparison, adverse Treasury QR comparison and later AR results show that rankings depend on the measurement process, horizon, history and model family. Fixed-report loss decomposition further distinguishes reductions in both mean error components from a score gain near an asymmetric-weight crossing. The estimators and supporting finite geometry are established operations; the comparison does not identify the causal value of additional features.

Later exploratory additions retain the original primaries and every adverse comparison. AR(1) residual-block replay qualifies interpretations based on QR alone; rank, floor and horizon-order checks clarify the issued reports; linear weight sensitivity concerns eight archived .95 reports. These additions use previously inspected results. The FX

evaluation archive had already been inspected before the conditional extension, and execution freezes do not restore an untouched holdout. The Treasury exercise uses a downloaded-vintage economic-date replay, rather than reconstructed historical public releases. Section 5 distinguishes the analysis stages and the supplement preserves their execution identities.

2. Closest work and established forecasting methods

Wisniewski, Lindsay and Lindsay (2020) already apply nearest-neighbor and other regression engines to this FX source family, using preceding positions and calendar or trading-session information. Their comparisons include quantile, conformal and normalized-conformal endpoint intervals. Chapter 6 of Wisniewski's thesis continues related regularized and normalized endpoint constructions (award year 2024; inspected cover 2023). Conditional regression, lag/calendar features, neighbors and normalization on these records are therefore precedents, rather than contributions of this study. Exact input-byte equality with the accepted version-2 member is not established.

Our response is a single upper quantile of an absolute multi-record maximum including the observed issue floor. A signed endpoint interval does not determine this peak: identical future-record marginals can yield different maxima when their dependence differs. The earlier paper averages two endpoint-quantile losses; its numbers cannot benchmark this single peak score. We instead refit a selected set of established engines to completed peak/headroom labels and evaluate the same response on common origins. This is not a reproduction of that earlier entire model suite.

Financial path extrema are established risk objects. MaxVaR studies the worst mark-to-market loss before a horizon under a portfolio-value first-passage framework (Boudoukh et al., 2004). Petrunia, Sabetti and Voia (2019, updated Discussion Paper 7) study the largest end-of-day net debit across ACSS participants, followed by block maxima or threshold excesses. Their cross-sectional terminal-debit maximum differs from a single series' future two-sided sampled peak. Van den Hoogen (2017) more directly defines an intraday liquidity buffer as a conditional quantile of the interval's maximum cumulative negative payment balance, using modeled transaction amounts and durations, simulated paths and EVT. Its directional cash-deficit target and confidential transaction inputs differ from this source-reported absolute position peak; that model is not reproduced here. These precedents rule out treating financial peak forecasting or a peak-to-quantile-buffer construction as new.

Prospective payment-system liquidity forecasting also has established precedents. Arjani, Li and Sabetti (2020) predict a remaining-day payment-demand/resource ratio and its threshold exceedance using payments and credit information. That ratio is a different functional from an absolute position peak or a directional timing peak. We neither reproduce their models nor observe the settlement inputs required to validate funding adequacy. The distinction further limits broad financial-target novelty claims.

Regression-quantile loss minimization is classical (Koenker and Bassett, 1978). Biau and Patra (2011) study sequential quantile prediction with lag-neighbor experts under stationarity, ergodicity and regularity assumptions. The fixed pools and delayed overlapping labels here do not inherit their consistency theorem. The comparison is deliberately bounded: quantile forests (Meinshausen, 2006), autoregressive quantiles such as CAViaR (Engle and Manganelli, 2004), sequential predictive conformal inference (Xu and Xie, 2023), and full volatility-updated or filtered historical simulation (Hull and White, 1998; Barone-Adesi et al., 1999) are established alternatives outside the executed menu.

Trajectory bands and financial horizon curves also predate this comparison. Conformalized Adaptive Forecasting of Heterogeneous Trajectories (Zhou, Lindemann and Sesia, 2024, arXiv v2) uses trajectory-level calibration and a cross-trajectory exchangeability premise. Sözen (2026) combines multi-horizon expected-shortfall forecasts, shape constraints and joint calibration. Their responses and assumptions differ from this position peak; an overlapping single archive is not automatically an exchangeable set of trajectories. Chronological slicing or response normalization alone transfers no simultaneous or conditional-coverage conclusion. We report realized loss, breaches and severity without borrowing those guarantees.

This study places selected mature methods on a common measured peak response, with a known floor, completed-label timing and construction/update controls. It provides a bounded empirical comparison rather than a new estimator. The directed literature search does not establish exhaustive priority for this exact target-and-data combination. The source evidence map records the consulted editions and selected reading ranges.

3. Sources, target and completed information

The FX input is the one-hour member of the Net Position Time Series Dataset, version 2: 18,576 records from 5 February 2014 01:00 through 24 January 2017 00:00. Wisniewski et al. (2020) describe aggregate client-flow positions across five liquid currency pairs for one market maker. We use datetime and the signed NetPosUsd scalar, rather than reconstructing raw currency positions, hedge transactions or cash movements. The source has 18,420 one-hour and 155 forty-nine-hour adjacent gaps. We preserve chronology without sorting, removing gap-crossing windows or interpolation. The timezone and historical publication/vintage convention remain unverified.

The Treasury source is New York Fed Primary Dealer Statistics series PDPOSGS-B, Treasury-bill long positions less short positions. FR 2004A supplies Wednesday-close positions at market/fair value. Its instructions use trade-date accounting, with specified buyback treatment on settlement date, and include eligible forward and when-issued outright positions; the series is not settled cash holdings (Board of Governors, March 2013 and July 2024 instructions). The accepted snapshot has 706 seven-day-spaced records from 3 April 2013 through 7 October 2026, in USD millions. It aggregates reporting dealers and can reflect valuation or reporting-composition changes. Taking an absolute value after summing signed positions permits cross-dealer cancellation: a small aggregate can coexist with large individual magnitudes (Duffie et al., 2023). It does not identify gross exposures or pure customer order flow.

$$B_t=|N_t|;\quad P_{t,H}=\max_{0\le h\le H}|N_{t+h}|;\quad G_{t,H}=P_{t,H}-B_t\ge 0.$$

The issue record is included, so B is a known response lower bound under the stipulated replay convention. H counts subsequent source records, and unobserved inter-record excursions are excluded. Six or 24 Treasury records span six or 24 weeks in this snapshot; FX calendar durations vary across its gaps. All horizons share the same source-specific evaluation issue origins, not the same final response dates.

$$r(t)=t_0+d\ \mathrm{floor}((t-t_0)/d);\quad T_{r,H,W}=\{r-H-W+1,\ldots,r-H\}.$$

At each refit r, every training origin satisfies $i+H\le r\le t$. The transform, response scale, coefficients or neighbor pool stay fixed until the next scheduled refit; query features and the known floor use actual issue t. Earlier evaluation labels enter subsequent fits only after maturity. Validation and evaluation restart their schedules at their declared segment starts. A frozen historical control freezes its pool, not its current query anchor.

Table 1. Frozen source-specific chronological budgets. All indices are zero based; W and d count source records. H is 6 or 24 in every cell.

Source	Validation origins	Evaluation origins	W	Refit d
FX	10463–11902 (1440)	11927–18551 (6625)	720 / 2880	24
Treasury	329–380 (52)	405–681 (277)	52 / 260	13

Treasury validation issue dates are 24 July 2019 to 15 July 2020. The final H=24 validation label ends at index 404 on 30 December 2020, before evaluation begins at 405 on 6 January 2021. The last common issue is 681 on 22 April 2026; its H=6 label ends at 687 on 3 June, and H=24 at 705 on 7 October 2026. FX validation ends at origin 11902; its latest H=24 label ends at 11926, before evaluation origin 11927.

Economic dates and public releases differ. New York Fed statistics normally update on Thursday for the preceding reporting week; a 2004 announcement established a one-week publication lag. Exact historical release times, holidays and original unrevised observations were not reconstructed. The inspected API routes supplied no verified historical release-vintage selector, which is not proof that no archive exists elsewhere. The Treasury exercise is a downloaded-vintage replay indexed by economic as-of date, treating the current record as known at issue by convention. It is not a live public-information or as-released backtest.

Strict loading accepts only the pinned source bytes, schemas, chronology and finite positions, without alias splicing. The accepted Treasury snapshot is archived with attribution and continuing source terms; FX redistribution permission remains unverified. The data guide and replication manifests give acquisition routes, exact hashes and byte counts. A hash identifies a downloaded version, not permanent availability. Changed downloads require a separately versioned replay rather than silent substitution.

4. Historical and conditional forecast procedures

4.1. Historical representations and update controls

For each completed episode i, retain its peak P, headroom G, lower increment l and upper increment u, including zero at h=0. At query signed anchor x=N(t), the translated peak is $V(i,t)=\max\{x+u(i),-x-l(i)\}$. For a common selected set A of size m, Q denotes the duplicate-preserving order statistic at j=ceil((m+1)19/20). The three reports are

$$F_t=B_t+Q(G_{i,H}); \quad S_t=B_t+Q(V_{i,t}-B_t); \quad K_t=\max\{B_t,Q(P_{i,H})\}.$$

Rolling controls take the latest W completed labels at each issue. Frozen controls use T(r,H,W), matching the conditional completed-label pool and refit schedule, while reanchoring S at current t. One-state controls select the nearest starting signed positions from the rolling eligible pool, resolving ties by recency: k=360 for FX W720, k=40 for Treasury W52 and k=80 for Treasury W260. No FX W2880 one-state reference was declared. The original construction and earlier recent/nearest menus remain historical evidence in Appendix B.

On a common set and fixed rank, let $r(\text{abs})=\max|B-|N(i)||$ and $r(\text{signed})=\max|x-N(i)|$. Elementary order-statistic and projection stability gives $|F-K|\le r(\text{abs})$, $|S-K|\le r(\text{signed})$, hence $|F-S|\le r(\text{abs})+r(\text{signed})$. Appendix A proves these finite relationships and their predictive limits. Nearest signed-state membership minimizes its maximum signed distance at a given count; it need not minimize forecast loss. These statements do not automatically bound QR, scaled responses or the richer 33-coordinate distance.

4.2. Predictors and training-only scaling

The 33 raw coordinates are N(t) through N(t−24), current magnitude, short/long past increment RMS, sine/cosine of recorded hour, sine/cosine of weekday (Monday zero), and whether the just-observed adjacent gap exceeds its nominal interval. RMS spans are 24/168 FX records and 6/26 Treasury records; nominal gaps are 3600 and 604800 seconds. Calendar coordinates use the stored source clock, not a verified UTC session. Weekly calendar coordinates can be constant, and coordinates need not supply independent information.

$$v_L(t)=[L^{-1}\,\mathrm{sum}_{j=t-L+1}^{t}(N_j-N_{j-1})^2]^{1/2}; \quad z_{i,r,j}=(x_{ij}-\mu_{rj})/s_{rj}.$$

Matching controls means the same W completed-response origins and refit schedule. Rich predictors can use earlier observations: at the oldest training origin i=r−H−W+1, long RMS reads N(i−168) for FX and N(i−26) for Treasury. This past-only lookback is not equalized across procedures.

Means and population standard deviations use completed training rows only. An exactly repeated column is centered at its repeated value and assigned scale one; a computed zero standard deviation also uses one. The same transform is applied to future queries within the refit block. There is no tuned small-variance threshold or whole-archive standardization. The positive monetary response scale is

$$a_r=\max\{\mathrm{mean}_{i\ \mathrm{in}\ T}(G_{i,H}),10^{-6}\,\mathrm{mean}_{i\ \mathrm{in}\ T}|N_i|,10^{-12}\}.$$

4.3. Ridge quantile and multivariate local reports

QR minimizes the normalized headroom regression-quantile objective with an unpenalized intercept and a ridge penalty on slopes. The fixed menu is lambda in {0.001,0.01,0.1}; tau=19/20. Training uses the unconstrained affine headroom prediction. Nonnegative projection is applied when issuing and validating a report, rather than inside the training objective; validation uses native units.

$$\min_{\beta_0,\beta} W^{-1}\,\mathrm{sum}_{i\ \mathrm{in}\ T}\,\rho_\tau(G_{i,H}/a_r-\beta_0-z_{i,r}^{T}\beta)+(\lambda/2)\|\beta\|^2.$$

$$C_t^{QR}=B_t+\max\{0,a_r(\hat{\beta}_{0,r}+z_{t,r}^{T}\hat{\beta}_r)\}.$$

This is not generally ordinary zero-centered ridge on the peak. The magnitude coordinate obeys B(i)=mu(B)+s(B)z(i,B). With alpha=mu(B)+a beta(0), theta=a beta+s(B)e(B), multiplying the headroom objective by a gives an equivalent unprojected peak objective with shifted penalty

$$W^{-1}\,\mathrm{sum}\,\rho_\tau(P_{i,H}-\alpha-z_{i,r}^{T}\theta)+[\lambda/(2a_r)]\|\theta-s_{rB}e_B\|^2.$$

This real-arithmetic parameterization includes the scale factor and floor-centered penalty; the intercept remains unpenalized. The issued report is max{B,alpha+z theta}. Adding an issue-known B translates conditional quantiles of

G into those of P. Raising a report below a known B≤P to B weakly reduces realized pinball loss. These elementary facts neither guarantee the fitted affine specification nor establish coverage.

For KNN, first divide each of the 25 training-standardized level coordinates by 5 and each coordinate of the RMS, hour and weekday pairs by sqrt(2); leave magnitude and gap unchanged. Compute squared Euclidean distances between the resulting coordinates. Exact distance ties favor more recent training origins, and selected paired rows retain chronological order. Plain KNN reports B+Q(k selected G). Scaled KNN divides each selected historical G by its own past short RMS, stabilized below by 10^{-6} a(r), and multiplies the resulting quantile by the correspondingly stabilized query RMS. The stabilizer is fixed within the refit block; no future query outcome or scale is used. This is response normalization, not a fitted dynamic path simulation.

$$s_{i|r}=\max\{v_{short}(i),10^{-6}a_r\};\quad C_t^{scaled}=B_t+s_{t|r}Q_k(\{G_{i,H}/s_{i|r}:i \text{ in } N_{t,r,k}\}).$$

The neighbor menus are {180,360,720} at both FX windows, {20,40,52} at Treasury W52, and {40,80,160} at W260. No interpolation is used; an unavailable corrected rank fails. For k=52, ceil((k+1).95)=51, while the ordinary .95 empirical pinball quantile has rank 50. At W720, W2880 and W260, however, W times .95 is integer: the lower ordinary rank and the adjacent corrected rank are both endpoints of the empirical pinball minimizer interval. Thus the W52 distinction does not imply that every corrected historical rule fails to minimize empirical pinball. A later endpoint/rank ablation is retained in Appendix D. Plain KNN at k=W equals frozen F exactly; its distances no longer affect membership. Scaled KNN still changes responses at k=W. Long-window menus do not contain the full pool. Appendix C records numerical acceptance and failures.

4.4. Separately frozen exploratory residual replay

After the conditional results had been inspected, we specified one additional path-propagation comparison without an order, volatility or tuning menu. At each original refit r, put a=r−H−W+1 and fit N(s)=c(r)+phi(r)N(s−1)+epsilon(s) on the unique transitions a+1,...,r. The W completed scenarios have W+H levels in their union and W+H−1 unique transitions; overlapping paths do not duplicate transitions in OLS. Coefficients and the historical pool remain fixed until the next original refit, while the query uses the actual N(t).

Using the same fitted coefficients for every residual in a block and replaying its original order from x=N(t) gives M(i,h;x)=N(i+h)+phi(r)^h[x−N(i)]. Set h0 explicitly to x and report the corrected empirical quantile of the W maximum absolute replay paths. The intercept cancels in this deterministic identity; it is still estimated in the mean recursion. At phi=1 in real arithmetic the arithmetic agrees with frozen S. Negative or above-one finite estimates are eligible, without clipping. Constant regressors, lost dispersion and nonfinite computations fail without a substitute model.

This is a mean-filtered ordered-block historical comparison, not the full conditional-volatility filtered simulation of Hull and White (1998) or Barone-Adesi et al. (1999). It does not standardize/rescale residuals, fit GARCH, independently draw future steps or introduce Monte Carlo randomness. Conditional representativeness of historical residual blocks remains unverified; numerical OLS acceptance supplies no coverage or mean-reversion law. Appendix A.4 gives the path relationship, and Appendix C records numerical checks.

5. Evaluation, scoring and analysis status

The FX primary is H6 QR W2880 minus original rolling F W720; the Treasury primary is H6 QR W260 minus frozen F W260. These are the two declared source-specific comparisons. The FX H24 QR W2880 minus original rolling S W720 comparison is a legacy-reference sensitivity: its reference was chosen in the preceding study, not independently of all historical outcomes. No favorable secondary model replaces either primary.

The wider comparison menu places those results in context. Every selected family is compared with frozen F/S/K at its own source/H/W (72 contrasts), and rolling-minus-frozen F/S/K supplies 24 update controls. The two FX contrasts add two, totaling 98 unique comparisons; the Treasury primary is already among the 72 and is not duplicated. The later AR addition has a separate exploratory comparison set, described below, and does not change this menu.

$$e=P-C;\quad L_{.95}=.95(e)_+ +.05(-e)_+;\quad D=M^{-1}\,\mathrm{sum}_t[L(P_t,C_t^{left})-L(P_t,C_t^{right})].$$

The stipulated pinball score weights an underprediction dollar nineteen times an overprediction dollar. It is a consistent quantile score (Gneiting, 2011), rather than an observed financial cost. We report U=mean$(P-C)_+$, the mean underprediction amount, and V=mean$(C-P)_+$, the mean overprediction amount. Both average over all issue origins; conditional breach severity averages over each method's own strict P>C event set. The reported headroom C−B is distinct from V. Archived fields mean_positive_overrun, mean_unused_allowance and mean_extra_allowance correspond to U, V and mean(C−B); their historical names do not represent resources actually allocated. Lower mean loss and fewer breach occasions are separate outcomes. Financial units are never pooled across sources.

Issue-year groups and the single globally anchored grid t(start)+j(H+1) are descriptive sensitivities. The grid never resets by year and removes shared recorded observations between retained response windows, while preserving shared training histories and possible time dependence. Counts are 947/265 in FX and 40/12 in Treasury for H6/H24. Different horizons retain different issue subsets. These groups are neither independent replications nor population confidence intervals. A partial year is retained and labeled as such. Every declared candidate must complete; failures stop its phase rather than produce a shortened mean, substitute method or favorable retry.

Table 2 records the complete validation-selected parameters. All 72 declared validation candidates completed. Parameter ties prefer the larger numeric candidate. Any failed or incomplete candidate makes its whole source/H/W/family cell ineligible; a failed selected evaluation has no complete loss, no shortened mean and no alternate-parameter fallback. Every actual declared phase completed.

Table 2. Complete validation-selected parameters. QR entries are ridge penalties; KNN entries are neighborhood counts. W52 plain KNN selects the full pool at both horizons.

Source	H	W	QR lambda	KNN k	Scaled k
FX	6	720	0.01	180	360
FX	6	2880	0.01	180	180
FX	24	720	0.01	180	180
FX	24	2880	0.001	180	180
Treasury	6	52	0.1	52	52
Treasury	6	260	0.001	80	80
Treasury	24	52	0.001	52	40
Treasury	24	260	0.1	160	160

The analysis has distinct stages. The original construction study fixed its target-specific procedure before calculating its forecasts and losses. The subsequent one-state retrieval extension was conceived after those outcomes were inspected; Appendix B retains both complete menus. The conditional extension fixed its sources, features, menus, machine limits and comparison definitions before its new validation runs, then recorded every selected parameter before evaluation. Its FX evaluation archive had already been inspected. These execution freezes establish what ran, rather than an unseen evaluation or removal of earlier inspection.

AR replay, floor/horizon diagnostics, ordinary-rank sensitivity, late conditional-breach evidence and fixed-report weight sensitivity were added after earlier results were inspected. Each is explicitly exploratory and leaves the original primaries unchanged. The public supplements retain their complete menus, design/freezing chronology, input identities, numerical acceptance conditions, independent arithmetic checks and reproduction receipts. Archive verification, numerical reproduction and new financial validation are distinct: the first two do not supply an additional sample.

6. FX results: conditional gains depend on history and horizon

Table 3. Complete conditional loss menu and same-history frozen controls on 6,625 FX origins, in USD millions. Every selected QR/KNN/scaled cell is shown. Rolling and one-state reference results and all breach/severity quantities remain in the complete CSVs.

H	W	F frozen	S frozen	K frozen	QR	KNN	Scaled KNN
6	720	0.442615	0.450774	0.932433	0.433246	0.436445	0.466292
6	2880	0.452693	0.450801	0.972162	0.385845	0.410583	0.426458
24	720	0.787416	0.755301	1.205808	1.035483	0.836210	0.896753
24	2880	0.763381	0.759371	1.240039	0.734695	0.769768	0.848004

The H6 primary QR W2880 loss is USD 385,844.89, against USD 441,774.04 for original rolling F W720. The direct paired mean is −USD 55,929.15, or −12.660%. Against W2880 frozen F it is −14.767%; against frozen S at that same window it is also favorable. Thus the added conditional benchmark changes the older FX ranking. The cross-window primary changes history length and update frequency as well as the estimator; the matched-pool controls address part, but not all, of this procedural difference. Figure 1 places every selected conditional cell beside its same-history frozen F reference.

The H6 gain is not uniform risk improvement. QR has 376 strict breaches (5.675%) versus 332 (5.011%) for old rolling F. Mean underprediction amount falls from USD 166,531.06 to 156,123.03, while mean overprediction amount falls by USD 920,830.41. More occasions breach, with a lower unconditional underprediction contribution. Conditional severity uses the different method-specific event sets and cannot identify improvement on the same incidents. At W720, H6 QR improves on frozen F by only 2.117%, while scaled KNN is worse by 5.349%. Figure 2 separates the score, breach-frequency and mean-error changes for the fixed headline comparisons.

At H24, QR W2880 loss is USD 734,695.07 against USD 748,383.54 for old rolling S W720: −USD 13,688.47 (−1.829%). Breaches increase from 287 (4.332%) to 531 (8.015%); mean underprediction amount rises from USD 149,813.88 to 317,678.12, while overprediction amount falls. With W720, QR instead loses to frozen F by 31.504% and frozen S by 37.095%. Neither the broader model class nor a favorable short-horizon result provides a uniform advantage across history budgets.

The H6 primary loss difference is negative in both issue-year groups: −12.012% over 6,262 origins in 2016 and −29.513% over the 363-origin partial 2017. The latter contains 5.48% of all issue origins and contributes 8.64% of the H6 primary net gain and 6.87% of the matched-history H6 net gain. For both H6 comparisons, U and V fall in each issue-year group and on the fixed 947-origin grid. In contrast, H24 improves against its legacy reference by only 0.555% in 2016 and 33.007% in partial 2017; 70.83% of its small total net gain comes from that short tail segment, and U increases in both groups and on the 265-origin grid. Figure 3 shows the issue-year contrasts. The fixed H+1 grids retain favorable H6/H24 legacy-reference loss differences of −16.388%/−8.027%. These are decompositions and sensitivities of the same inspected record, not independent validation or a basis for retrospective forecast switching.

7. Treasury results: adverse QR comparisons

Table 4. Complete selected conditional loss menu and same-history frozen controls on 277 Treasury origins. Losses are in the source's USD-million units, not FX USD. H6/H24 mean six/24 weekly transitions. No monetary average is pooled across sources.

H	W	F frozen	S frozen	K frozen	QR	KNN	Scaled KNN
6	52	2779.054	2786.620	2574.098	5095.519	2779.054	4208.293
6	260	2758.665	2707.244	2287.758	4159.673	2663.561	3127.173
24	52	3715.693	3652.379	5017.741	12165.574	3715.693	5625.667
24	260	2480.306	2478.871	2301.476	5204.760	2257.963	4137.264

The declared H6 QR W260 primary has mean loss 4,159.673 versus 2,758.665 for matched frozen F, in USD millions. Its paired increase is 1,401.008 (+50.786%), with 54 rather than 28 strict breaches (19.495% versus

10.108%). Mean underprediction amount rises from 1,395.639 to 3,131.862 and overprediction amount falls from 28,656.166 to 23,688.088. Under the stipulated score, the reduced overprediction amount does not compensate for the more heavily weighted underprediction increase. This is an adverse accuracy result from numerically accepted fits, not a convergence failure.

Every Treasury QR cell loses to its own frozen F control: +83.354% and +50.786% at H6 for W52/W260, and +227.411% and +109.843% at H24. Short-history H24 QR breaches 132 of 277 responses. All four selected scaled-KNN cells also have higher loss than their matched F controls. The primary QR difference is adverse in each complete issue-year group from 2021 through 2025. Its favorable −14.404% in 2026 covers only 16 issue origins through April, with no breaches for either method. This partial period is retained, not used to rescue the primary.

Plain KNN W52 selects k=52 at both horizons and equals frozen F at every stored issue. The distance coordinates do not change membership in these rows; This is an ordinary validation-selected full-pool identity, rather than a response to fitting failure or evidence of a feature-membership gain. At W260, selected KNN has secondary all-origin loss reductions of 3.447% (H6) and 8.964% (H24) against F. These do not replace the QR primary, and no selected conditional procedure is descriptively below every retained historical reference at its horizon. Such a whole-menu observation is not a newly selected forecast rule.

The secondary H24 KNN W260 comparison has 13 versus 12 breaches and mean underprediction amount 327.816 versus 306.848, alongside reduced overprediction amount. On the single globally anchored 12-origin H+1 grid its mean loss is 3,111.788 against 2,807.400 for F, reversing to +10.842%. The H6 KNN sensitivity does not reverse: its 40-origin difference remains −22.071%. These exact grids remove shared recorded label observations but retain common training histories and possible time dependence. The sparse reversal does not identify a cause or prove an opposite population ranking; it does prevent a claim of a stable secondary ranking.

All 24 same-source conditional cells and all 66 reference procedures completed. The complete source-specific results, all 72 common-history contrasts, 24 update controls, annual decompositions and sparse-grid contrasts remain available. The adverse Treasury primary and short-history results are not omitted or replaced by the best hindsight-selected configuration.

8. Exploratory dynamic replay: family-specific score and tail tradeoffs

Table 5. Complete later AR comparison. Entries are within-reference mean-loss percentages; negative favors AR. S/F have matching source, H, W and refit schedule; QR is the original selected same-cell fit. Counts are AR / S strict breaches over 6,625 FX or 277 Treasury origins. Every row is exploratory; sources are not pooled.

Source	H	W	AR−S %	AR−F %	AR−QR %	Breaches AR / S
FX	6	720	-1.536	+0.278	+2.447	282 / 255
FX	6	2880	-0.546	-0.962	+16.197	338 / 268
FX	24	720	+2.386	-1.790	-25.318	401 / 289
FX	24	2880	+5.147	+4.595	+8.679	444 / 292
Treasury	6	52	-13.444	-13.208	-52.665	22 / 11
Treasury	6	260	-18.887	-20.399	-47.209	31 / 28
Treasury	24	52	+35.125	+32.823	-59.432	56 / 33
Treasury	24	260	-24.118	-24.162	-63.860	59 / 12

AR has lower loss in five of eight cells against S, five against F and five against QR; these sets differ. Relative to QR it loses in three of four FX cells and improves in all four Treasury cells. The original adverse Treasury QR primary therefore does not establish failure of every conditional family. Against S, the FX H6 changes are −1.536%/−0.546% at W720/W2880, while H24 changes are +2.386%/+5.147%. Treasury changes are −13.444%/−18.887% at H6 and +35.125%/−24.118% at H24 for W52/W260. No additional model is selected from this table.

Against S, AR mean report and mean overprediction amount decrease in every cell, while strict breach occasions increase in all eight and mean underprediction amount increases in seven. Treasury H24 W260 loss falls from

2478.871 to 1881.016 USD millions (−24.118%), yet breaches rise from 12 to 59 of 277 and mean underprediction amount rises from 306.848 to 892.437. Mean overprediction amount falls from 43747.307 to 20664.031. Under the stipulated score, .05 times the overprediction reduction exceeds .95 times the added underprediction. This explains the observed loss change without asserting safer limits, calibrated tails or observed financing savings.

Direct-S signs reverse for FX H24 W720 on the 265-origin grid (+2.386% overall, −1.884% sparse) and Treasury H24 W52 on only 12 origins (+35.125%, −14.100%). Treasury H24 W260 remains favorable on its sparse grid but is adverse in 2022 and 2023 (+9.928%, +20.224%). Every annual and sparse contrast remains. Observed fitted coefficients lie between zero and one, with refit means around .981–.985 for FX and .676–.850 for Treasury; these descriptions do not identify true mean reversion. Source, horizon, history and model family all matter, and neither average report reduction nor a below-one slope determines a loss ranking.

9. Loss-weight sensitivity of eight fixed reports

The main score fixes an underprediction/overprediction weight ratio of 19:1. We now vary that ratio for previously issued .95 reports. This changes the evaluation of fixed actions, rather than fitting other quantiles. For report m, U and V are its mean underprediction and overprediction amounts over the same origins. For 0<=a<=1 define:

$$J_m(a)=a\,U_m+(1-a)V_m;\ r=a/(1-a);\ J_m(.95)=L_{m,.95}.$$

At a=0 or a=1 only one component remains; finite positive r corresponds to 0<a<1. For reports A and B the difference is $a(U_A-U_B)+(1-a)(V_A-V_B)$. Reducing both components gives weak improvement for every constant nonnegative linear weighting, and strict improvement when a positively weighted component strictly decreases. Opposing component differences yield a crossing when it lies in the stated range. This elementary geometry establishes neither dominance under nonlinear or state-dependent costs nor dominance over every consistent .95 quantile score (Gneiting, 2011). A higher breach frequency remains possible when U falls.

Table 6. Preference sensitivity of the four named fixed comparisons. Differences are left minus right. FX amounts are divided by 10^6; Treasury amounts already have USD-million source units. The sources and economic horizons are not pooled. r is the stipulated shortfall/excess weight ratio; equality holds at a finite displayed crossing, before rounding. Reference definitions remain those of Section 5.

Comparison	dU	dV	Crossing r	Left preferred
FX H6 primary	-0.010408	-0.920830	None	All r >= 0
FX H6 matched	-0.029549	-0.775527	None	All r >= 0
FX H24 legacy	+0.167864	-3.463190	20.630898	r < 20.630898
Treasury H6 primary	+1736.222894	-4968.077943	2.861429	r < 2.861429

The two FX H6 comparisons reduce both mean underprediction amount and mean overprediction amount. Their improvement therefore does not rely on the single 19:1 preference, even though strict breach occasions still increase. By contrast, the FX H24 legacy comparison increases U while reducing overprediction amount: its 1.829% pinball improvement reverses at r=20.630898 (a=0.953770), close to 19. The Treasury H6 primary becomes preferable only below r=2.861429 (a=0.741028); its adverse result at 19 remains intact. These are descriptive consequences of fixed sample means, not estimates of financing prices or risk preferences.

Within each source/H/W cell, the lower envelope is computed only over eight archived .95 reports: QR, KNN, scaled KNN, matched frozen F/S/K, AR and I. They share the completed-label history length and refit schedule, but this finite menu does not exhaust feasible reports. For example, C=B is feasible and has V=0, making it a minimizer at a=0 over all feasible reports; that floor-only action is absent from this eight-report menu. The reported endpoint winners therefore describe the menu, rather than all feasible decisions. Rolling and one-state reports are also excluded; the original cross-window primary references are separately labeled in Table 6. Appendix E retains all full and sparse-grid envelopes, endpoints and ties. Full-grid Treasury H6 W260 QR is never optimal among these eight reports, including both endpoints. At a=.95 the menu minimizer changes on four of eight sparse grids: FX H24 W720, Treasury H6 W52/W260 and Treasury H24 W52. Treasury H24 grids contain only 12 origins. These are dependent within-archive sensitivities, rather than four independent replication outcomes.

This analysis was conceived after earlier results and preliminary preference calculations were inspected. It reads saved aggregate means without accessing raw observations or refitting reports. The supplement preserves the before-run execution identities, both byte-identical result runs and the strengthened independent affine-inequality check. Rational calculations are exact for the saved decimal mean tokens, which summarize floating-point losses rather than infinite-precision raw observations. The resulting hindsight envelopes are descriptive objects, not deployment selectors.

10. Interpretation, coherence and remaining limits

The principal finding is variation in ranking. Long-history QR improves the six-record FX comparison and performs worse on the separately fitted Treasury aggregate. Exploratory AR replay improves on QR in all four Treasury cells and loses in three of four FX cells. Against matched S, AR improves mean loss in five cells and raises it in three, while breach occasions increase in all eight. Thus the adverse Treasury QR primary concerns the tested QR specifications, and the favorable AR cells do not establish a general advantage for conditional modeling.

Mean quantile loss and breach frequency answer different questions. The two H6 FX comparisons reduce U and V despite more strict breaches; their sample-average improvement persists under every constant nonnegative linear weighting of those components. Higher breach frequency rules out empirical first-order stochastic dominance in the favorable direction for the distribution of positive-part errors $(P-C)_+$, including zeros: its empirical distribution function is lower at zero. A cost per breach or a nonlinear loss can therefore change the ranking. The smaller H24 FX legacy gain instead trades increased U for reduced V and reverses at ratio 20.63. These are properties of fixed reports under stipulated weights, rather than measured financing costs or uniformly improved tail risk.

The controls address only part of the procedural difference. F/K already use current magnitude and S uses the signed anchor; QR and local methods also change function class, regularization, local count, scaling, validation selection and raw lookback. Matching completed-response pools and refit schedules does not isolate a causal information effect. Each source is fitted and validated separately; fitted FX coefficients and transforms are not transferred to Treasury. The source primaries also differ in references, histories, update schedules, units and calendar horizons.

Limited validation budgets further restrict generalization. Five of eight QR selections lie at penalty-menu endpoints, including all four Treasury cells; this does not show that an untested penalty would help. The 52 consecutive Treasury validation origins contain at most three mutually observation-disjoint H24 label windows, without implying an effective sample size of three. Issue-year groups and sparse grids remain subdivisions of the inspected archives. They cannot replace compatible untouched financial records, and mature forecasting alternatives remain outside the tested menu.

Post-result diagnostics clarify the accepted reports without repairing them. The ordinary empirical-rank control retains the matched H6 QR improvement in FX and deterioration in Treasury. Separately fitted horizon reports can cross even though every realized target is ordered; numerical acceptance supplies no horizon-coherence guarantee. A binding floor is not an identified cause of the Treasury QR errors, and projection onto a known outcome floor differs from raising a long-horizon report to another learned report. Appendix D retains complete rank, floor and crossing results. Its late fixed-terminal conditional-breach calculations remain exploratory and provide no retrospective coverage certification for the selected analysis.

The remaining limits concern information, measurement and decisions. The Treasury economic-date replay does not reconstruct historical public releases or revisions, and its aggregate can conceal offsetting dealer positions. Both sources omit inter-record paths, book decomposition and the quotes, fills, hedges, cash flows, collateral and financing prices needed for an operational evaluation. Active inventory-sensitive quoting, as in Avellaneda and Stoikov (2008), changes the position process and is outside this passive exercise. Further compatible data, a defensible release clock and an observed decision link would answer stronger questions than another comparison on the same archives.

11. Conclusion

Established methods change the ranking of historical forecasts for the sampled net-position peak, with results depending on source, horizon, history and model family. The declared six-record FX QR primary lowers mean loss by 12.66%, with a 14.77% matched-history reduction and more breach occasions. The separately fitted six-week Treasury QR primary raises loss by 50.79% and increases breaches. Later AR replay improves on QR in every Treasury cell while increasing breaches relative to matched increment translation in all eight cells. Fixed-report weight sensitivity separates the short-horizon FX reductions in both mean error components from the near-threshold long-horizon gain; sparse-grid reversals further limit claims of stable rankings. The retained evidence supports a bounded retrospective benchmark using mature methods. It establishes neither a new estimator, conditional coverage nor an observed benefit from changing trading, funding or risk-limit decisions.

Appendix A. Finite geometry and its predictive limits

A.1. Common-set projection bounds

These construction arguments concern F/S/K on an identical historical set and rank. They are retained from the earlier comparison. A one-dimensional signed-distance minimax property does not extend automatically to the multivariate neighbor distance, and the bounds do not automatically constrain QR or scale-adjusted responses.

These are elementary finite-array arguments, rather than new asymptotic or coverage results. Fix a nonempty common selected set J, an available rank $1\le k\le|J|$, query $x=N_t$, floor $B=|x|$ and completed historical windows. All identities in this appendix concern real arithmetic. Write $B_i=|N_i|$, $d_{i,h}=N_{i+h}-N_i$, with $d_{i,0}=0$, and define

Let a(i)=min over h of d(i,h) and b(i)=max over h of d(i,h), so $a(i)\le 0\le b(i)$. These are the lower and upper increments denoted l and u in Section 4. The auxiliary nonnegative envelope coordinates below use u(i)=b(i) and v(i)=−a(i).

$$A_i=B+P_i-B_i;\quad T_i=\max_{0\le h\le H}|x+d_{i,h}|;\quad C_B(z)=\max(B,z).$$

Both A_i and T_i are at least B. The projection C_B is nondecreasing and 1-Lipschitz. It commutes with a finite order statistic, as does a common translation. Hence $F_J=Q_k(A)$, $S_J=Q_k(T)$ and $K_J=Q_k(C_B(P))$. Since $C_B(A_i)=A_i$,

$$|A_i-C_B(P_i)|\le|A_i-P_i|=|B-B_i|.$$

The reverse triangle inequality bounds each path value's change under a starting-position translation by $|x-N_i|$. Taking a maximum gives $|T_i-P_i|\le|x-N_i|$. Applying the same floor projection preserves that bound because $T_i\ge B$. Finally, paired order statistics are nonexpansive in the maximum coordinate difference: if $|u_i-v_i|\le r$ for every i, then $u_i\le v_i+r$ and conversely; monotonicity and common translation give $|Q_k(u)-Q_k(v)|\le r$. These observations prove the two K bounds in Section 4. The triangle inequality and $||N_i|-|x||\le|N_i-x|$ prove the stated F/S bound. Keeping the rank fixed while changing one side's historical rows does not by itself justify these common-set comparison bounds.

Among all size-m subsets, selecting the m smallest signed distances minimizes the largest selected distance: any alternative must include an element at least as far away as the mth ordered distance. This finite minimax fact, with the declared tie convention, concerns the signed radius. It does not minimize the absolute-anchor radius, the sum of the two radii, the actual report gap or a forecast loss. For every fixed realized y, pinball loss is $\max(\tau,1-\tau)$-Lipschitz in its report, so a report-gap bound also bounds the absolute loss difference. Such stability supplies no sign for the realized or expected loss difference.

A.2. Envelope plateaus and the role of signed anchors

Put $u_i=b_i\ge 0$ and $v_i=-a_i\ge 0$. The headroom of an increment envelope at any anchor z is

$$g_i(z)=\max(z+u_i,-z+v_i)-|z|;\ \text{for } z\ge 0 \text{ it equals } \max(u_i,v_i-2z),\ \text{and for } z\le 0 \text{ it equals } \max(u_i+2z,v_i).$$

F pools each envelope's headroom at its historical anchor, $g_i(N_i)$; S pools its headroom at the query, $g_i(x)$. The branches agree at zero and together make g_i 2-Lipschitz, with range lying in $[\min(u_i,v_i),\max(u_i,v_i)]$. Apply the distance and range bounds to each paired row, then use the same-rank order-statistic stability from A.1 to obtain

$$|F_J-S_J|\le\max_{i\text{ in }J}\min\{2|x-N_i|,|u_i-v_i|\}.$$

For example, $u_i=v_i$ in every row makes $F_J=S_J$ at arbitrary anchors. Nonnegative query/source anchors together with $u_i\geq v_i$ in every row also suffice; the nonpositive analogue uses $v_i\geq u_i$. These are sufficient conditions, not a characterization: pooled order statistics can coincide even when paired rows differ. The range bound also gives $F_J,S_J\leq R_J$, where $R_J=B+Q_k\{\max(u_i,v_i):i \text{ in } J\}$. This is a theoretical same-set ordering; no R retrieval configuration is evaluated in the extension, and the ordering does not make R_J preferable under a given loss.

If every $B_i\leq B$, then $A_i\geq C_B(P_i)$ and $F_J\geq K_J$. If every $B_i\geq B$, then $C_B(P_i)=P_i\geq A_i$ and $F_J\leq K_J$. Equal absolute anchors force F=K but leave signs relevant to S. A coherent full-pool example uses c>0, H=1, n=720, τ=19/20, t=721 and $N_s=(-1)^{s+1}c$ for $0\leq s\leq 721$. The pool uses completed labels at origins J={1,...,720}, with 360 anchors of each sign and current $N_t=c$. All native peaks equal c and headrooms equal zero, so F=K=c. Raw reanchoring gives 360 peaks c and 360 peaks 3c; rank 685 gives S=3c. Thus $r_{abs}=0$ while $r_{signed}=2c$ and $|S-K|=2c$. This artificial path respects maturity and an available corrected rank; it is not an empirical claim or a claim about which rows nearest retrieval selects.

A.3. The same observed history allows opposite loss rankings

Choose d>0, H≥1, $0<\tau<1$ and n large enough that $k=\text{ceil}((n+1)\tau)\leq n$. Set t=H+n−1 and $N_s=d(t-s)$ for $0\leq s\leq t$, using J={0,...,n−1}. Every selected response ends by t. Each historical peak is its own positive starting value, so $G_i=0$ and F=0. Each increment path equals −dh; reanchoring at $N_t=0$ gives S=Hd. Two continuations of this identical observed history yield:

Table A1. Artificial coherent continuations. Symbols are constructed quantities, not source-record estimates.

Future N(t+h), 1≤h≤H	Actual peak	F loss	S loss
0	0	0	(1−τ)Hd
−dh	Hd	τHd	0

Either report can therefore have strictly smaller realized loss under a compatible future. A report ordering also need not hold universally. Consider the artificial history (0,−4,−1,2), with H=1, n=3, τ=3/4, issue t=3 and J={0,1,2}. The current anchor and floor are x=B=2; all three historical windows are completed. Table A2 shows the native and raw-reanchored paths.

Table A2. Completed H=1 windows from one artificial history, in arbitrary position units. P is the native peak, G the native headroom and T the peak after reanchoring at x=2. These values are not source-record estimates.

i	Historical pair	P	G	Reanchored pair	T
0	0, −4	4	4	2, −2	2
1	−4, −1	4	0	2, 5	5
2	−1, 2	2	1	2, 5	5

The available corrected rank is ceil((3+1)3/4)=3. The third order statistics of G, T−B and P are 4, 3 and 4, respectively, so F=6, S=5 and K=4. The small pool and τ=3/4 belong only to this illustration. It contains no observation after issue and gives no realized query loss. These constructions establish logical limits, not the conditional law of the recorded series or a population ranking. A deterministic construction can refute an unconditional mathematical assertion while supplying no empirical validation for either rule.

A.4. Raw translation versus fixed-coefficient residual replay

For interpretation, fix arbitrary coefficients c and phi and define residuals epsilon(s)=N(s+1)−c−phi N(s). Replay a historical window's ordered residuals from query x. The coefficients are fixed after a refit; this algebra does not assert a true dynamic law or a martingale-difference property. Denote replay by M and raw translation by W. Subtracting the recursions and using induction gives

$$M_{i,h}(x)=N_{i+h}+\varphi^h(x-N_i);\quad W_{i,h}(x)=N_{i+h}+(x-N_i).$$

Taking the maximum absolute path value yields

$$|\max_h|W_{i,h}(x)|-\max_h|M_{i,h}(x)||\leq\max_{0\leq h\leq H}|1-\varphi^h|\,|x-N_i|.$$

For $0 \le \varphi \le 1$ the coefficient is $1-\varphi^H$; at $\varphi=1$ the paths coincide. The identity needs no residual law, while treating past residuals as representative of future conditional residuals is a separate assumption. Section 8 fits a specified mean recursion and compares realized loss, but does not validate that conditional residual-representativeness assumption. The extrema suffice to evaluate the peak after raw translation, but generally not after the dynamic correction, which depends on path order. The same-rank reports inherit a deterministic bound with max over historical anchor distances, not a guaranteed sign of loss improvement. Conditional validity remains a separate question.

Appendix B. Earlier construction and retrieval evidence

This appendix preserves the complete loss menus of the 21 October historical studies. Monetary displays are source-reported FX USD millions. The original rules include headroom F, raw reanchoring S, a symmetric increment envelope R, native peak K (original L), and the current-floor report P0. One factor changes from the original H6/W720/tau=.95 specification in each sensitivity cell. All use the same 6,625 issue origins; different preferences are not pooled into an overall winner.

Table B1. Complete original mean loss grid. Δ is the stored paired mean F minus S; negative favors F within that setting.

Setting	F	S	R	K	P0	Δ F−S
main	0.441774	0.451149	0.527544	0.925885	1.813620	-0.009375
h12	0.592518	0.604781	0.689604	1.058614	2.950594	-0.012263
h24	0.777069	0.748384	0.803106	1.197833	4.630827	+0.028685
n480	0.449843	0.460816	0.532515	0.885243	1.813620	-0.010973
n1440	0.443793	0.448275	0.526979	0.931562	1.813620	-0.004482
tau90	0.659643	0.659499	0.763481	1.477201	1.718166	+0.000144
tau99	0.154991	0.158998	0.175486	0.288834	1.889983	-0.004007

The earlier retrieval extension takes all 720 completed origins, the most recent 360/180, or the nearest signed-state 360/180 from that eligible pool, with recency ties. Corrected ranks for 720/360/180 are 685/343/172. Its H6 nearest360 K-minus-recent360 K primary lowers loss by 31.83% against a weak native-level baseline, while both original full720 F and S remain lower. This prior-menu leadership statement does not extend to the new conditional models in Sections 6–7.

Table B2. Complete earlier retrieval-extension mean loss menu and same-set F-minus-S paired mean, in source-reported USD millions. Full720 rows repeat original references. Differences are computed before display rounding.

H	Historical set	F loss	S loss	K loss	F−S mean
6	full720	0.441774	0.451149	0.925885	-0.009375
6	recent360	0.454089	0.467678	0.872516	-0.013589
6	nearest360	0.448664	0.448731	0.594811	-0.000067
6	recent180	0.485866	0.478940	0.785878	+0.006926
6	nearest180	0.468133	0.459592	0.525471	+0.008541
24	full720	0.777069	0.748384	1.197833	+0.028685
24	recent360	0.807496	0.824047	1.246630	-0.016551
24	nearest360	0.791192	0.770354	0.960696	+0.020838
24	recent180	0.858214	0.895718	1.284092	-0.037504
24	nearest180	0.847734	0.827305	0.963931	+0.020429

All 12 fixed nearest-minus-recent score contrasts were negative. Each matched S gain accompanied more breaches and greater mean underprediction amount; conditional severities describe method-specific event sets. January 2016 was adverse for the primary K retrieval comparison at both horizons. The original primary F-minus-S mean was

−USD 9,374.86 (−2.08%), with 332 versus 254 breaches and greater mean underprediction amount; its H24 and tau=.90 rankings reversed. The latter tau=.90 difference was small and did not establish equivalence.

The original diagnostic groups, monthly outcomes and four stationary-bootstrap-style profiles remain in results/diagnostics.csv and stability.csv. The fixed resampling used 999 circular geometric-block replicates per mean length 120,240,720,1440, PCG64 SeedSequence([20261020,length]), and noninterpolated order statistics 25/975. The stationarity, moment and dependence premises behind Politis and Romano (1994) were not established here; those profiles are mechanism sensitivities, not population confidence intervals, and are not applied to the new conditional contrasts. The earlier extension's complete 30-row summary, 22 paired comparisons, 10 geometry rows and 26 monthly display rows remain under results/revision-20261021/. No prior adverse evidence is removed.

Appendix C. Numerical, rank and reproduction conventions

All issued reports use stored float64 values. Historical ranks retain repeated responses without interpolation. An unavailable rank stops. The corrected ranks for k=20,40,52,80,160,180,360,720 are 20,39,51,77,153,172,343,685. A corrected rank is not always an empirical pinball minimizer: k=52 selects rank51 instead of the ordinary .95 rank50. At integral k tau, the upper adjacent order statistic is an endpoint of the real-arithmetic empirical minimizer interval. Neither fact establishes future coverage.

For S, the translated peak is computed first, current B is subtracted before ranking, then B is added. Final losses use direct stored error P−C, not an algebraically equivalent differently rounded headroom residual. The fitter evaluates its complementary weight as 1−tau; final scoring converts 19/20 and 1/20 separately to floats. Display rounding never defines a breach, tie or model selection. Null conditional severities and relative differences retain explicit denominator reasons.

QR implements mature two-block convex ADMM (Boyd et al., 2011), with 34 coefficients, an unpenalized intercept, penalty rho=1/W, at most 10000 iterations, absolute tolerance 10^{-6} and relative tolerance 10^{-5}. Training checks finite values and linear-system solve residuals. Acceptance requires both primal and dual ADMM residual thresholds and a separate feasible-dual objective-gap condition. These numerical checks do not certify a statistical forecasting law. Each fit initializes slopes at zero and the intercept at the inverted-CDF training quantile, following the frozen implementation. An objective-gap bound does not bound issued-report differences or ranking sensitivity: with an unpenalized intercept the pinball objective can be flat or nearly flat. For example, at real tau=19/20, an intercept-only sample of nineteen zeros and one M>0 has every intercept in [0,M] as a minimizer. This illustrates the certificate's scope; it does not assert that the accepted financial fits exhibited that example or exact nonuniqueness.

$$P_{lambda}=\text{mean}\ \rho_{\tau}(y-Xb)+(\lambda/2)\|\beta\|^2;\quad D_{lambda}(v)=\text{mean}(vy)-\|Z^{T}v/W\|^2/(2\lambda).$$

Here y=G/a, X includes an intercept and Z its standardized slopes. The candidate dual vector satisfies tau−1≤nu≤tau and exact zero sum on an inward 2^{-40} dyadic grid; integer box limits use the exact binary-float representation of tau. The checksum checks the represented vector exactly, while objectives remain floating-point evaluations. A nonnegative Fenchel/stationarity decomposition evaluates the gap without subtracting similar objectives; the objective difference must agree within 256 machine epsilons times max(1,|P|,|D|), and the stable gap must be at most $10^{-6}(1+|P|)$. This is not interval-arithmetic certification.

All 72 validation candidates and 24 selected evaluations passed their declared numerical/resource conditions; 768 validation and 1196 evaluation QR refits were accepted. Public receipts retain scalar diagnostics, coefficient/dual-vector hashes and maturity metadata. The original outcome audit used private forecast arrays and direct source-window maxima. Analytic and independently enumerated convex fixtures, future perturbations and explicit failures support the implementation checks; these checks do not establish model adequacy or a forecasting law.

One financial worker and one requested numerical thread ran at a time, under declared time and memory budgets. The result supplements retain actual elapsed times, process high-water RSS and observed thread snapshots, separately for calculation and checking. These observations are neither continuous OS hard limits nor research hours.

The public result package retains every phase receipt, validation candidate, procedure summary, annual subdivision, declared contrast and sparse-grid contrast. Historical checker revisions and their original receipts remain available.

Arithmetic verification reads the exact hashed archive bytes and binds parameter labels and complete inventories; it does not create a new financial sample.

Exact numerical reproduction requires the pinned source snapshots and frozen code/specification, not merely a current download or the same series label. Source routes may change their bytes through revision or expansion. New phase receipts have new timestamps/resources and therefore different hashes even if numerical forecasts agree; they must not overwrite archived evidence. The shared structured paper content generates TeX and the separately rendered supplied PDF; document generation reads approved text and aggregate-only figures, not raw financial inputs.

The public reproduction route separates read-only archive verification from explicit financial recomputation. With the pinned inputs, reproduce_conditional.py replicate creates a new attempt and serially reruns all 56 phases. The recorded replay matched the 24 selected cells, all 72 validation-candidate scalar rows and all six aggregate CSV bytes. A separate diagnostic replay matched the floor and horizon tables without another fit. New timestamps and resources need not match old receipt hashes. Commands and measured scopes are in the repository guide; the source versions are the same, so this is numerical reproduction rather than independent financial validation.

AR uses a common finite power-of-two scale and anchor-centered OLS cross-products with normal-equation checks. Its supplement retains the pre-execution synthetic challenges, complete eight-cell run and a separate scalar/Fraction readback of all aggregate rows, 24 selected exact OLS fits and 24 scalar recursive reports. The selected fit/report oracles do not independently reconstruct every financial fit. Full implementation history and resource observations remain outside the scientific comparison.

Appendix D. Explicitly post-result report diagnostics

After viewing and publishing the complete primary results, we recorded a descriptive floor-binding/horizon-coherence plan and evaluated the already accepted forecasts. These checks are not part of the 98 before-execution contrast menu, and they modify no forecast or tuning choice. The original 90 configurations and 45 source/window/method pairs are retained in post-result-floor.csv and post-result-horizon.csv, with hashes and process resources in their separate audit. Every common realized target satisfies P(H6)≤P(H24). Those files cover the original QR/KNN/scaled-KNN study and its historical controls. The later AR diagnostics are separate files; the original tables and forecasts are unchanged.

Table D1. Original QR/KNN/scaled-KNN horizon-coherence diagnostics; the later AR addition is not included. A crossing means stored C(H6)>C(H24), with no tolerance used to redefine the event. Maximum gap is displayed in USD millions (FX divided by 10^6; Treasury already in that unit). The distinct financial quantities are not pooled.

Source	W	Method	Crossings	Origins	Cross %	Max gap
FX	720	QR	640	6625	9.660	18.742
FX	720	KNN	22	6625	0.332	1.930
FX	720	SCALED_KNN	48	6625	0.725	1.775
FX	2880	QR	140	6625	2.113	5.303
FX	2880	KNN	0	6625	0.000	0.000
FX	2880	SCALED_KNN	18	6625	0.272	1.772
Treasury	52	QR	91	277	32.852	139288.627
Treasury	52	KNN	39	277	14.079	8202.000
Treasury	52	SCALED_KNN	72	277	25.993	124615.389
Treasury	260	QR	80	277	28.881	79417.646
Treasury	260	KNN	0	277	0.000	0.000
Treasury	260	SCALED_KNN	25	277	9.025	18835.763

Historical short Treasury controls also cross: rolling F/S on 38/37 issues, frozen F/S on 39/39, and one-state F/S/K on 20/20/3. Treasury W260 one-state K crosses on 13; its other historical controls and all listed FX historical controls

have zero recorded crossings. Their H-specific mature sets differ, so common-set construction monotonicity is no guarantee across these procedures. The two selected k=W52 KNN reports equal their F counterparts, including the 39 crossings.

Exact floor equality describes a stored report, not an observed negative latent prediction. Treasury H24 QR W52 binds on 65 issues (58 breaches), with another 74 breaches off the floor; W260 binds on three (two breaches), with 82 others. The realized zero-headroom outcomes are 14/277 at H24 and 50/277 at H6. FX has 662/6,625 and 1,523/6,625 respectively. These descriptive event sets do not identify the cause of misspecification or effective independent sample size.

The separate AR addition retains all eight floor configurations and four same-source/window horizon pairs. FX W720/W2880 have no observed crossings over 6,625 origins each; Treasury W52/W260 have 59/26 over 277 (21.300%/9.386%), with maximum positive gaps 11784.473/5347.870 USD millions. Every realized target is ordered. Treasury W52 AR binds its floor on seven issues at each horizon, with five H6 and seven H24 breaches. The other six AR configurations never bind. Those empty-group breach rates are undefined, not zero. These inspected-sample diagnostics establish neither a cause nor a coverage guarantee.

The report-order problem does not by itself justify a particular repair. For example, B=0, P(H6)=P(H24)=1, C(H6)=10 and C(H24)=1 satisfy the target order. Raising the long report to the short one changes its .95 loss from zero to .45. This artificial real-arithmetic example is not source data; it distinguishes a learned report from a known lower bound. No isotonic reconciliation, horizon switching or post-result parameter search was run.

The original AR diagnostic command remains restricted to its retained manifest-bound attempt. The separately demonstrated fresh-attempt route chains run_ar1_replay.py, verify_ar1_output.py and reproduce_ar1_diagnostics.py with new identifiers and the genuinely generated version-2 readback identity. It reproduced all six AR aggregate CSVs and both diagnostic tables byte-for-byte without rerunning the conditional phases. The guide results/revision-20261022/ar1-fresh-reproduction/README.md gives the actual commands and verification scope. This is same-source reproduction, rather than additional financial validation.

A later exploratory diagnostic uses the bounded strict hit I=1{P>C}, with equality a nonbreach, and the null that its conditional expectation given full issue information is at most alpha=1/20 at every declared issue. It is not unconditional equal-loss testing or exact exceedance identification at a floor atom. Conditional-moment betting and delayed binary evidence are mature precedents (Casgrain et al., 2023, v2; Henzi and Ziegel, 2022, v3). For a fixed alternative u in [alpha,1], the hit/no-hit factor is u/alpha or (1−u)/(1−alpha); its null conditional expectation is at most one. Every globally anchored H+1 phase retains only issues whose previous same-phase target has resolved.

$$E_{n,k} = [1/(1-\alpha)] \int_{\alpha}^{1} (u/\alpha)^{k} [(1-u)/(1-\alpha)]^{n-k} \, du.$$

$$E_{cell} = [1/(H+1)] \sum_{r=0}^{H} E_{n(r),k(r)}; \ \alpha=1/20.$$

The prior mixes each whole phase product, then all phases receive equal weight; empty phases have value one. Iterated expectation and nonnegative averaging justify a fixed-terminal e-value for a genuinely fixed causal procedure under the stated null, without independence, stationarity or a financial-magnitude moment assumption. This gives no full-filtration supermartingale or arbitrary-stopping claim. The complete matched family has 56 labels, including aliases, and excludes 24 rolling and 18 one-state reports. Its formal reference threshold is 56/(1/20)=1120. Because this diagnostic and family were chosen after inspecting results, the crossings do not obtain retrospective empirical five-percent familywise error control. The fixed-procedure proof also does not establish individually calibrated e-values or p-values for this same-sample, post-selected analysis: selection need not be measurable at the original issue times. No prior, phase or menu was retuned.

*Table D2. All 56 late fixed-terminal evidence diagnostics, rounded to three significant figures; * marks an exact crossing of the fixed 1120 reference threshold. TB is the separately fitted Treasury aggregate; F/S/K are the matched frozen controls. A noncrossing is not coverage certification, and evidence magnitude is not a financial-quality ranking. All 896 phase rows and exact rational identities are retained in the supplement.*

Source H/W	QR	KNN	Scaled	F	S	K	AR
FX 6/720	104	0.00905	0.00476	0.0112	0.00397	26.5	0.00527
FX 6/2880	0.0378	0.00923	0.00476	0.0183	0.00441	294	0.0125
FX 24/720	8.5e7*	1.18	0.171	0.146	0.0164	270	0.0558
FX 24/2880	0.734	0.153	0.0486	0.0318	0.0159	5.12	0.146
TB 6/52	1.31e8*	0.057	0.0537	0.057	0.057	0.38	0.242
TB 6/260	2.2e3*	0.637	0.305	0.919	0.919	0.401	1.95
TB 24/52	3.81e4*	19.7	0.949	19.7	19.7	5.18	2.46
TB 24/260	59.4	0.202	0.13	0.221	0.221	4.53	11

The four reference crossings are FX H24 W720 QR and Treasury H6 W52/W260 and H24 W52 QR. Treasury H24 W260 AR has 59/277 strict hits and evidence 11.0284, below 1120; this does not negate its adverse breach record. Two Treasury QR cells both have 84/277 hits but different evidence: about 1.311×10^{8} at H6 W52 and 59.4235 at H24 W260. Their phase sizes, hit allocation and forecasting tasks differ. No power, equivalence or financial-quality conclusion follows. A distinct integer-formula readback reconciled every hit, phase and exact rational value without raw sources, refitting or report repair. Complete inputs, proof and reproduction scope are in results/revision-20261022/coverage-evalues/README.md.

Ordinary empirical-rank ablation

After inspecting both source archives and the previous results, we fixed one further eight-cell supplement and published a 16-file execution freeze before its financial calculation. At each original evaluation-segment refit, the added intercept-only report uses the same W completed headrooms and original update schedule as frozen F. It selects the lower empirical .95 pinball quantile with duplicate preservation and no interpolation, then adds the actual current magnitude. There is no new tuning, selected model, source, evaluation subset or changed primary.

$$C^{I}_{t} = B_{t} + G_{(\mathrm{ceil}(19W/20)),r,H}; C^{F}_{t} = B_{t} + G_{(\mathrm{ceil}(19(W+1)/20)),r,H}.$$

Here G(j),r,H denotes the jth ordered headroom in the original mature pool. For the six cells with W720, W2880 or W260, the two ranks are endpoints of the same ordinary empirical pinball minimizer interval. Only the two W52 cells have noninteger W times tau, yielding ordinary rank50 versus corrected rank51; ties can collapse distinctions in general. This is an endpoint/rank sensitivity, not a claim that corrected F is universally non-ERM. Dividing all headrooms by a positive training-response scale and multiplying the fitted intercept back by that scale preserves the rank and the minimizer interval in native units.

Table D3. Complete post-inspection ordinary empirical-rank ablation. Percentages are QR-minus-I and I-minus-F mean-loss changes relative to the right report; negative favors the left. QR means and counts are imported unchanged from the original selected fits. I/F counts are newly reconstructed; columns list I/F/QR strict breaches over 6,625 FX or 277 TB origins. H counts source records. Sources are not pooled.

Source	H	W	Ranks I/F	QR−I (%)	I−F (%)	Breaches I/F/QR
FX	6	720	684/685	−1.967	−0.153	337/332/531
FX	6	2880	2736/2737	−14.773	+0.008	365/360/376
FX	24	720	684/685	+31.508	−0.003	432/427/924
FX	24	2880	2736/2737	−3.783	+0.026	359/357/531
TB	6	52	50/51	+87.086	−1.995	14/11/84
TB	6	260	247/248	+46.058	+3.237	32/28/54
TB	24	52	50/51	+208.258	+6.213	37/33/132
TB	24	260	247/248	+105.020	+2.353	15/12/84

All eight reconstructed F summaries match every original summary metric exactly. The lower report I is strictly below F at every issue in this archive; its breach count and mean underprediction amount increase in all eight cells, while mean overprediction amount decreases. Its mean loss improves in only three cells. For example, Treasury H24 W52 I raises loss by 6.213% against F despite using the lower ordinary empirical minimizer. QR remains worse than I in all four Treasury cells, while the matched FX H6 W2880 improvement is 14.773%. The ablation supports these finite signs and tradeoffs under the stipulated convention, without removing the inspected-sample, model-universe or measurement limits.

The supplement retains all I/F summaries, annual and globally anchored sparse-grid rows, and 24 contrasts under results/revision-20261022/rank-ablation/. QR-minus-I uses the original unrounded QR mean and the added I mean, rather than a new originwise reconstruction of QR traces. Direct I-minus-F paired differences are retained separately. An independent scalar-window/Python-sorting implementation reconstructed every I/F report and reconciled all targets, scores and aggregate tables. It shares the strict source parser and accepted data, so this is independent arithmetic rather than external financial validation. Actual resource observations and check counts remain in the supplement; no earlier fit or adverse result was replaced.

Appendix E. Lower envelopes of eight archived reports

Each row follows a from 0 to 1 through its maximal open intervals. NN=KNN; Sc=scaled KNN; F/S/K=matched frozen historical controls; AR=AR residual replay; I=ordinary empirical headroom quantile. A slash separates consecutive intervals. Adjacent displayed intervals tie at their common exact boundary; equals joins identical minimizing lines. Breakpoints are rounded for display. The public results.json retains exact rational boundaries and every endpoint/tie set, all 128 input summaries, 448 pair relations and coefficientwise dominance. Endpoints are evaluated separately and are not inferred from rounded interior thresholds. Ratios r=a/(1-a) should be converted from the exact fractions, especially near 1.

Table E1. All eight full-origin frontiers among the same eight fixed reports; 6,625 FX or 277 Treasury origins per row. Units are source native and are never pooled.

Source	H	W	Preferred sequence	Interior a boundaries	At .95
FX	6	720	QR / NN / S	0.952882, 0.964505	QR
FX	6	2880	QR / Sc	0.972817	QR
FX	24	720	QR / NN / AR / S	0.845488, 0.889246, 0.938369	S
FX	24	2880	QR / S	0.957033	QR
Treasury	6	52	QR / AR / NN=F	0.717896, 0.972846	AR
Treasury	6	260	AR / K	0.960158	AR
Treasury	24	52	QR / K / S	0.423917, 0.897882	S
Treasury	24	260	K / AR / NN / S	0.778365, 0.970017, 0.995666	AR

Table E2. All eight existing globally anchored H+1 sparse-grid frontiers. Counts are FX 947/265 and Treasury 40/12 for H6/H24. These are descriptive sensitivities with shared training histories.

Source	H	W	Preferred sequence	Interior a boundaries	At .95
FX	6	720	QR / NN / Sc	0.950621, 0.984383	QR
FX	6	2880	QR / Sc	0.978149	QR
FX	24	720	QR / NN / AR / S	0.805398, 0.910344, 0.959437	AR
FX	24	2880	QR / S	0.966600	QR
Treasury	6	52	QR / K / NN=F	0.645837, 0.995462	K
Treasury	6	260	AR / K	0.937124	K
Treasury	24	52	QR / K / AR / NN=F=S	0.693667, 0.940131, 0.990484	AR
Treasury	24	260	K / AR	0.721694	AR

For full Treasury H6 W52, NN and F have identical component means and share their final region. For sparse Treasury H24 W52, NN, F and S share the final region. The complete mathematical comparison retains a report that is optimal only at a boundary even if it has no open interval; absence from a displayed sequence alone is not an endpoint claim. The .95 winner changes in four cells on the sparse grid. The lower envelope within this inspected eight-report menu is a hindsight descriptive object, not an estimated optimal decision rule, an independence argument or a confidence set. As in Section 9, the omitted feasible floor-only report prevents interpreting these endpoint or interval minimizers as an optimization over all feasible actions.

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Figure captions

Figure 1. All 24 selected conditional forecasts versus frozen F with the same source, horizon, history window and refit schedule. Points show within-reference mean-loss percentages; negative favors the conditional procedure. Source panels have explicit scales and are not pooled. Full-pool KNN identities are marked. No sampling interval or conditional-coverage claim is shown.

Figure 2. The two declared source primaries and FX H24 legacy-reference sensitivity: relative mean-loss, underprediction and overprediction changes, plus breach-rate differences in percentage points. The preserved figure labels mean positive overrun and mean unused allowance denote U and V, respectively. These are descriptive mean errors of passive reports, rather than combined utility or observed monetary savings.

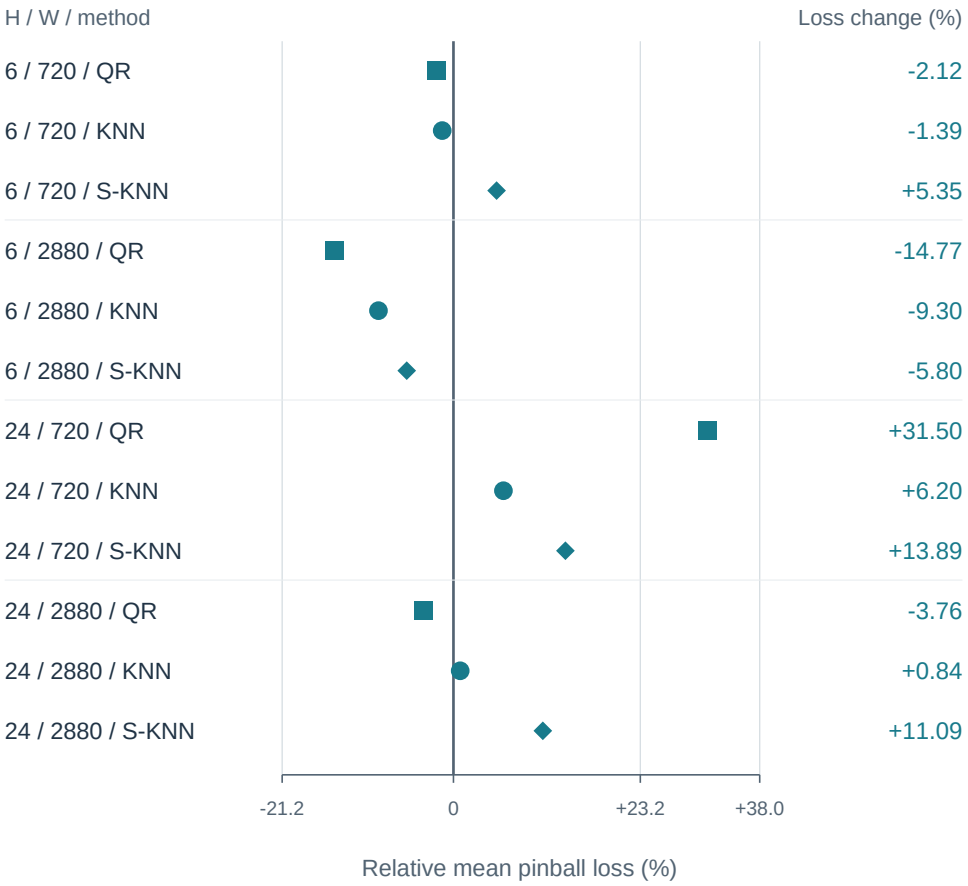
Figure 3. Every issue-year group for the two declared primaries and FX H24 legacy sensitivity. Values are paired within-reference loss percentages, with origin counts and partial years labeled. They are subdivisions of the same source records, not independent replications, confidence intervals or retrospectively selected switching regimes.

Conditional forecasts against matched historical headrooms

Mean loss change (%) relative to own W/H frozen F | negative is lower loss

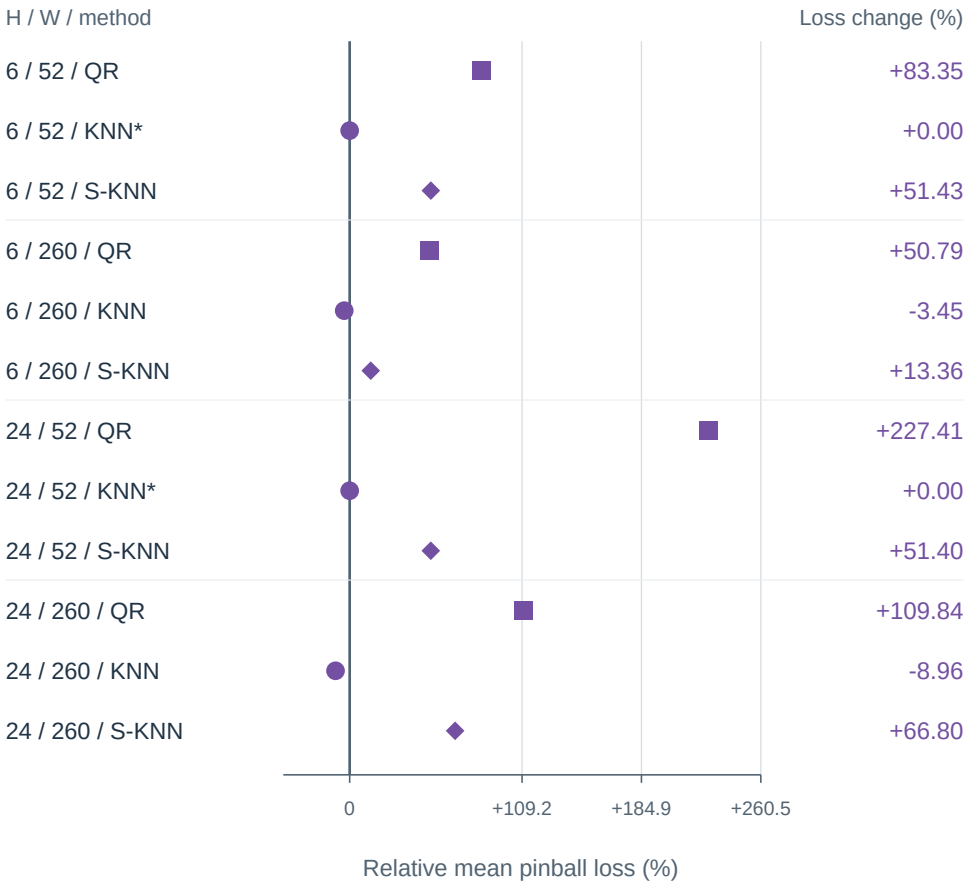
FX client-flow archive

Native: USD | N=6625 | 12 configurations



Treasury-bill dealer aggregate

Native: USD millions | N=277 | 12 configurations



QR: ridge quantile regression | KNN: local headrooms | S-KNN: scale-adjusted local headrooms

* KNN uses the full pool: exact 0.00% at NY W=52 for both H; these rows show no feature-membership gain.

H counts records. Each panel has its own axis range; no native monetary totals are combined.

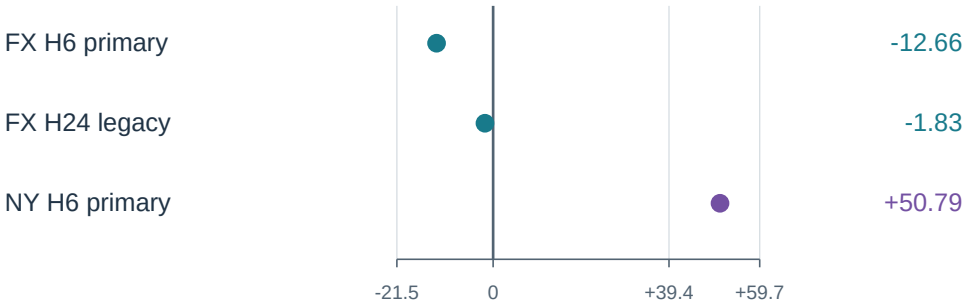
Descriptive, overlapping response windows. No confidence intervals; the FX interval was previously inspected.

Source: accepted evaluation-audit-02 exports | all selected conditional configurations retained

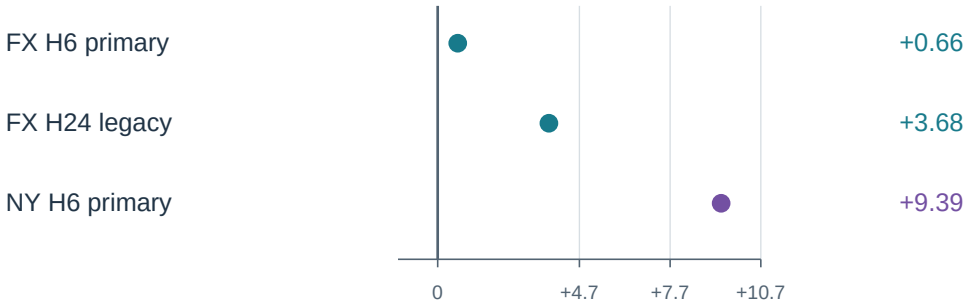
Three fixed comparisons: loss and breach tradeoffs

Each point is QR minus its stated reference | aggregate-relative changes; no native-dollar pooling

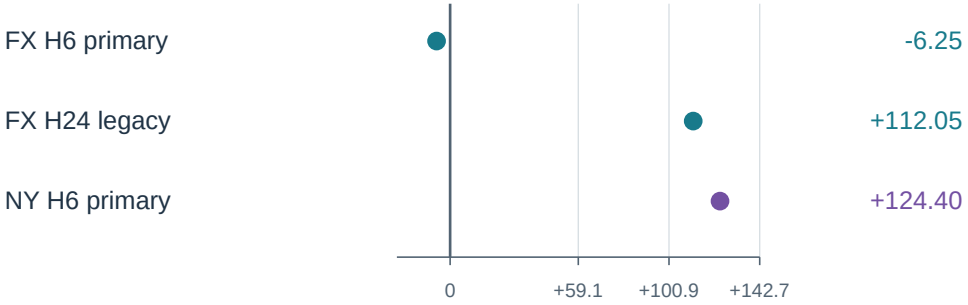
Mean pinball loss change (%)



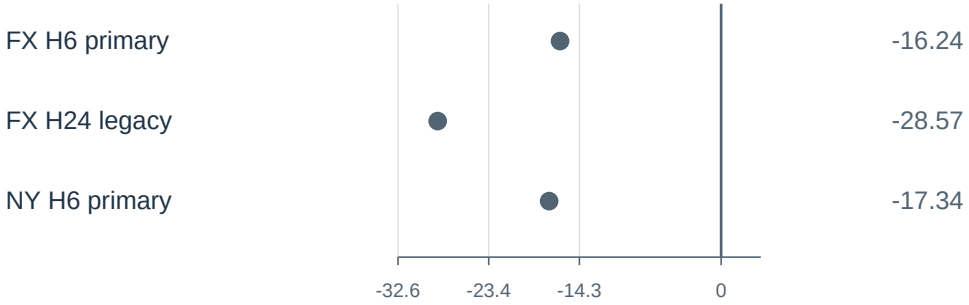
Strict breach rate (percentage points)



Mean positive overrun change (%)



Mean unused allowance change (%)



FX H6 primary: QR W=2880 vs rolling F W=720 | N=6625

FX H24 legacy: QR W=2880 vs rolling S W=720 | N=6625

NY H6 primary: QR W=260 vs frozen F W=260 | N=277

Relative (%) = 100*(QR-reference)/reference; breach change uses the same paired N in percentage points.

Mean overrun averages max(P-C,0) across all origins. Unused sign is descriptive; no preference coloring.

Descriptive, overlapping windows. No confidence intervals; a lower score is not an operational saving.

Source: accepted paired-comparisons.csv plus matching evaluation-summary.csv aggregates

Annual loss contrasts for the same three comparisons

Relative mean loss (%) | negative is lower loss | annual origin counts shown; partial years retained

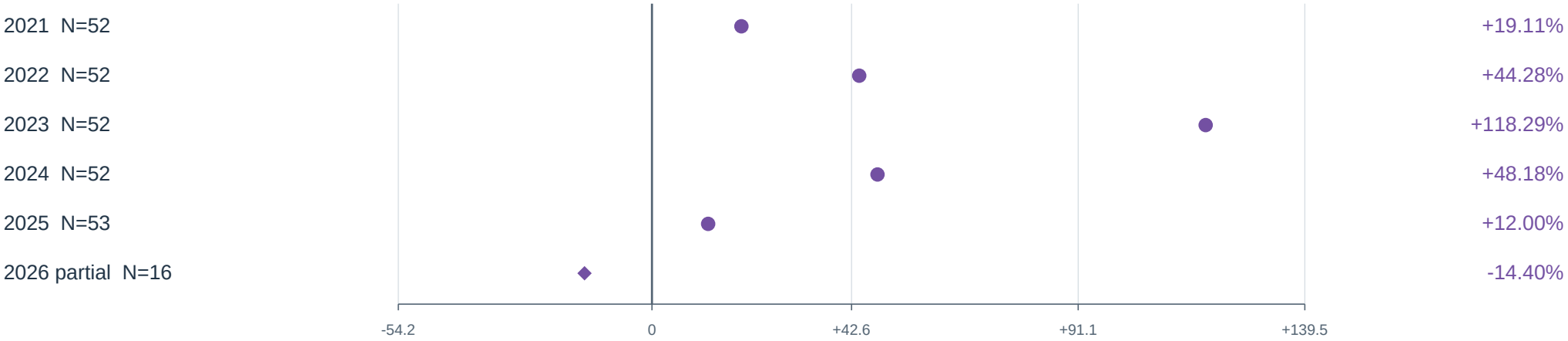
FX H6 PRIMARY | QR W=2880 vs rolling F W=720



FX H24 LEGACY | QR W=2880 vs rolling S W=720



NY H6 PRIMARY | QR W=260 vs frozen F W=260



Shared percent axis; source-native units differ. Annual ratios are computed within each stated contrast.

N counts overlapping origins, not independent replicates. No confidence intervals or error bars.

Source: accepted paired-annual.csv | diamonds mark 2017 / 2026 partial years; no years are omitted